Pingbang Hu

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Education

- **University of Illinois Urbana-Champaign** — **Illinois**
  Ph.D. Candidate in Information Science, School of Information Science — *Aug. 2023–Present*
  - **Advisor**: Jiaqi Ma
- **University of Illinois Urbana-Champaign** — **Illinois**
  M.S. in Applied Mathematics, College of Liberal Arts & Sciences — *Aug. 2023–Aug. 2025*
  - **Concentration**: Optimization and Algorithms
- **University of Michigan** — **Michigan**
  B.S. in Computer Science with **Summa Cum Laude**, *College of Engineering* — *Aug. 2021–May 2023*
  - **Minor**: Mathematics, College of Literature, Science, and the Arts
- **Shanghai Jiao Tong University** — **Shanghai, China**
  B.E. in Electrical and Computer Engineering, UM-SJTU Joint Institute — *Aug. 2019–Aug. 2023*
  - **Minor**: Computer Science, UM-SJTU Joint Institute

Research and Industry Experience

- **Hudson River Trading** — **New York**
  Algorithm Developer Intern — *Jan. 2027*
- **Deep Learning Research, Susquehanna International Group** — **Pennsylvania**
  Machine Learning Ph.D. Intern — *June 2026–Aug. 2026*
- **ML Security Team, Anthropic** — **San Francisco**
  AI Safety Research Fellow — *Jan. 2026–May 2026*
- **AWS AI Lab, Amazon** — **New York**
  Applied Scientist Intern — *May 2025–Aug. 2025*
- **Sugiyama Laboratory, National Institute of Informatics** — **Tokyo, Japan**
  Research Intern — *May 2024–Aug. 2024*
- **SURE Program, University of Michigan** — **Michigan**
  Undergraduate Researcher — *May 2022–Apr. 2023*

Peer-Reviewed Conference Publications

(* denotes equal contribution)

[C1] **Pingbang Hu**, Mahito Sugiyama, "Pseudo-Nonlinear Data Augmentation: A Constrained Energy Minimization Viewpoint". In *Proceedings of the 14th International Conference on Learning Representations* (ICLR 2026)

[C2] **Pingbang Hu**, Joseph Melkonian, Weijing Tang, Han Zhao, Jiaqi W. Ma, "GraSS: Scalable Data Attribution with Gradient Sparsification and Sparse Projection". In *Proceedings of the 39th Advances in Neural Information Processing Systems* (NeurIPS 2025)

[C3] Yiwen Tu*, **Pingbang Hu***, Jiaqi W. Ma, "A Reliable Cryptographic Framework for Empirical Machine Unlearning Evaluation". In *Proceedings of the 39th Advances in Neural Information Processing Systems* (NeurIPS 2025)

[C4] Xinhe Wang, **Pingbang Hu**, Junwei Deng, Jiaqi W. Ma, "Adversarial Attacks on Data Attribution". In *Proceedings of the 13th International Conference on Learning Representations* (ICLR 2025)

[C5] Yuzheng Hu, **Pingbang Hu**, Han Zhao, Jiaqi W. Ma, "Most Influential Subset Selection: Challenges, Promises, and Beyond". In *Proceedings of the 38th Advances in Neural Information Processing Systems* (NeurIPS 2024)

[C6] Junwei Deng*, Ting-Wei Li*, Shiyuan Zhang, Yijun Pan, Hao Huang, Xinhe Wang, **Pingbang Hu**, Xingjian Zhang, Jiaqi W. Ma, "`dattri`: A Library for Efficient Data Attribution". In *Proceedings of the 38th Advances in Neural Information Processing Systems Datasets and Benchmarks Track* (NeurIPS 2024) (**Spotlight**)

Peer-Reviewed Journal Publications

(* denotes equal contribution)

[J1] Junwei Deng*, Yuzheng Hu*, **Pingbang Hu***, Ting-Wei Li*, Shixuan Liu*, et al., "A Survey of Data Attribution: Methods, Applications, and Evaluation in the Era of Generative AI". In *Journal of Artificial Intelligence Research* (JAIR 2026)

Workshop Papers

(* denotes equal contribution)

[W1] Shixuan Liu, Tongli Zhou, Junwei Deng, **Pingbang Hu**, Jiaqi W. Ma, "`dattri-LLM`: A Unified and Efficient Library for Training Data Attribution at LLM Scale". In *3rd Workshop on Attributing Model Behavior at Scale: Data Attribution and Provenance* (NeurIPS 2026)

[W2] **Pingbang Hu**, Xueshen Liu, Z. Morley Mao, Jiaqi W. Ma, "**Dr. Post-Training**: A **D**ata **R**egularization Perspective on LLM Post-Training". In *Workshop on Scientific Understanding of Foundation Models* (COLM 2026)

[W3] Junwei Deng, **Pingbang Hu**, Suliang Jin, Hao Lu, Jiachen T. Wang, Shichang Zhang, Jiaqi W. Ma, "How Faithful Is Trajectory-Based Data Attribution? Error Sources, Remedies, and Practical Guidelines". In *Workshop on Actionable Interpretability Workshop* (COLM 2026) (**Oral**)

[W4] **Pingbang Hu**, Yuzheng Hu, Jiaqi W. Ma, Han Zhao, "A Unified Theory of Random Projection for Influence Functions". In *3rd Workshop on Navigating and Addressing Data Problems for Foundation Models* (ICLR 2026)

Pre-prints and Technical Reports

(* denotes equal contribution)

[P1] **Pingbang Hu***, Xinyan Hu*, Chloe Loughridge, Xuanli He, Nicholas Carlini, Matthew Jagielski, Javier Rando, Yiming Zhang, "Agentic Backdoors via Pretraining Poisoning". *Preprint*

Teaching Experience

- **Graduate Teaching Assistant, University of Illinois Urbana-Champaign** — **Illinois**
  Hold discussion and office hours weekly, design assignments and exam problems, grade and guide projects.
  - **Network Analysis**: A graduate-level course on the M.S. IS track. — *Spring 2025, Spring 2026*
- **Instructional Aide, University of Michigan** — **Michigan**
  Hold discussion and office hours weekly, design assignments and exam problems, grade and guide projects.
  - **Introduction to Cryptography**: An upper-level course on the main undergraduate CS track. — *Winter 2023*
  - **Randomness and Computation**: A graduate-level course on the M.S. CS theory track. — *Fall 2022*
- **Teaching Assistant, Shanghai Jiao Tong University** — **Shanghai, China**
  Hold discussion and office hours weekly, design and grade assignments and exams.
  - **Honor Mathematics III**: An undergraduate-level course on the main B.Eng. ECE track. — *Summer 2021*
    - **Competition**: Hold the 1st UM-SJTU JIntegration Bee competition.
  - **Honor Mathematics II**: An undergraduate-level course on the main B.Eng. ECE track. — *Fall 2020*

Invited Talks

- **Agentic Backdoor via Pre-Training Poisoning**
  - UK AI Security Institute — *Nov. 2026*
  - Anthropic — *May 2026*
- **Data-Centric AI: Faithful Data Attribution and Data Regularization**
  - Google DeepMind — *Aug. 2026*
- **Towards Market Data Valuation under Complex Training**
  - Susquehanna International Group — *Aug. 2026*
- **Science of Data: Predictable, Optimizable, and Scalable**
  - Citadel GQS — *May 2026*

Honors and Awards

- **Anthropic AI Safety Research Fellowship** — **San Francisco**
  Fellows (32 out of 2000+ applicants worldwide) for AI Safety Research at Anthropic — *Oct. 2025*
- **Graduate Conference Travel Award** — **Illinois**
  Graduate College's Competition at University of Illinois Urbana-Champaign — *Nov. 2024*
- **NeurIPS 2024 Scholar Award** — **British Columbia, Canada**
  Financial Aid Award for NeurIPS 2024 — *Oct. 2024*
- **Excellent Internship Award** — **Tokyo, Japan**
  Excellent (best) internship evaluation at National Institute of Informatics — *Aug. 2024*

- **Hong Kong, Macao and Taiwan Overseas Chinese Student Scholarship** — **Shanghai, China**
  First Prize (Ranked #2) among all HK, MC, and TW students at Shanghai Jiao Tong University — *Oct. 2021*
- **Undergraduate Excellent Scholarship** — **Shanghai, China**
  Third Prize among all students at UM-SJTU Joint Institute — *Nov. 2020*
- **Bao Gang Excellent Scholarship** — **Shanghai, China**
  Second Prize (Ranked #3) among all Taiwan students at Shanghai Jiao Tong University — *June 2020*
- **Hong Kong, Macao and Taiwan Overseas Chinese Student Scholarship** — **Shanghai, China**
  First Prize (Ranked #1) among all HK, MC, and TW students at UM-SJTU Joint Institute — *Dec. 2019*

Professional Service

- **Workshop Organizer**
  The 3rd Workshop on Regulatable ML @NeurIPS 2025
- **Program Committee**
  AAAI 2027, AAAI 2025
- **Conference Reviewer**
  ICLR 2027, NeurIPS 2026, ICML 2026, ICLR 2026, NeurIPS 2025, ICLR 2025, ICML 2024, IEEE BigData 2023
- **Journal Reviewer**
  TMLR, ACM Computing Surveys